

ITSAR2 313 – SYSTEM INTEGRATION AND ARCHITECTURE 2

LABORATORY REPORT: EXERCISE 2 — MICROSERVICES EDGE CASE TESTING

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Abstract

This laboratory exercise tested three Express.js-based microservices—student-service (port 4001), course-service (port 4002), and enrollment-service (port 4003)—against a 27-step sequential test suite covering HTTP status codes 200, 201, 400, 404, 409, 503, and 504. All tests were executed using Windows PowerShell and curl.exe -i to capture both response headers and JSON bodies in a deterministic order, ensuring each step had the required data before proceeding to the next. The key finding is that the enrollment-service correctly identifies and handles external HTTP dependency failures, successfully returning 503 (Service Unavailable) for offline dependencies and 504 (Gateway Timeout) for slow dependencies. All 27 test steps were confirmed to return the correct HTTP status codes and consistent error messages across all services.

I. INTRODUCTION

Edge case testing verifies that a system handles abnormal inputs and failure conditions correctly. Testing only the "happy path"—valid inputs producing expected outputs—is insufficient for production systems. Real-world APIs receive malformed requests, reference nonexistent resources, and depend on services that can fail or respond slowly. Each of these scenarios must return a meaningful HTTP status code and a structured error response that clients can parse and handle programmatically.

HTTP status codes communicate the outcome of every API request:

- **2xx codes** indicate success: 200 OK for reads and updates, 201 Created for new resources.
- **4xx codes** indicate client errors: 400 Bad Request for invalid input, 404 Not Found for missing resources, 409 Conflict for duplicate entries.
- **5xx codes** indicate server-side failures: 503 Service Unavailable when a dependency is offline, 504 Gateway Timeout when a dependency is too slow.

Microservices introduce 503 and 504 scenarios that monolithic systems never face. Because a monolith has no external HTTP dependencies, connection failures and timeouts are architecturally impossible.

The test suite in this report is ordered sequentially so that each step builds on the data created by previous steps. This deterministic approach ensures reproducible results and eliminates test failures caused by missing or inconsistent data. A fresh database reset is performed before every full run to establish a known baseline of four member records, two default courses, and two default enrollments.

II. METHODOLOGY/IMPLEMENTATION

All three services were started in separate PowerShell terminal windows before testing began using the `node src/server.js` command. The enrollment-service depends on the student-service and course-service for cross-service validation, so all three must be running simultaneously. A fourth terminal was used exclusively for `curl.exe` commands.

The following table lists every edge case scenario implemented and tested in the Express.js environment:

Status	Error Response	Trigger	Service
201	<i>N/A (Success)</i>	Successful creation	All
200	<i>N/A (Success)</i>	Successful read / update / delete	All
400	<code>{"error": "... are required"}</code>	Missing or invalid input fields	All
404	<code>{"error": "[Resource] not found"}</code>	Resource ID does not exist	All
404	<code>{"error": "Student does not exist"}</code>	Invalid studentId on enroll	Enrollment

404	{"error": "Course does not exist"}	Invalid courseId on enroll	Enrollment
409	{"error": "Student email already exists"}	Email already registered	Student
409	{"error": "Student is already enrolled..."}	Student already in course	Enrollment
503	{"error": "Student service unavailable"}	Dependency service offline	Enrollment
504	{"error": "Student service unavailable (timeout)"}	Dependency response too slow	Enrollment

A. Reset and Pre-conditions

Before starting, SQLite databases (students.db, courses.db, enrollments.db) and previous text evidence files were removed to ensure a clean state. The backend automatically seeds initial dummy data representing the team members upon startup.

```

● PS C:\Users\Sean\Desktop\SAR2> cd C:\Users\Sean\Desktop\SAR2\Lab2
PS C:\Users\Sean\Desktop\SAR2\Lab2> Remove-Item .\services\student-service\students.db -ErrorAction SilentlyContinue
● PS C:\Users\Sean\Desktop\SAR2\Lab2> Remove-Item .\services\course-service\courses.db -ErrorAction SilentlyContinue
● PS C:\Users\Sean\Desktop\SAR2\Lab2> Remove-Item .\services\enrollment-service\enrollments.db -ErrorAction SilentlyContinue
PS C:\Users\Sean\Desktop\SAR2\Lab2> Remove-Item .\docs\evidence\*.txt -ErrorAction SilentlyContinue
● PS C:\Users\Sean\Desktop\SAR2\Lab2>

```

Baseline after reset:

- students: pre-seeded with 4 member records
- courses: pre-seeded with 2 default records
- enrollments: pre-seeded with 2 default records

Default seeded IDs used by the sequence:

- students: IDs 1-4 (member records), new test student becomes ID 5
- courses: IDs 1-2 (default records), new test course becomes ID 3
- enrollments: IDs 1-2 (default records), new test enrollment becomes ID 3

B. Start Services Open four separate terminal window

Terminal 1:

```
PS C:\Users\Sean\Desktop\SAR2> ...
PS C:\Users\Sean\Desktop\SAR2\Lab2\services\student-service> node src/server
.js
Student Service running on http://localhost:4001
```

Terminal 2

```
PS C:\Users\Sean\Desktop\SAR2> cd C:\Users\Sean\Desktop\SAR2\Lab2\services\c
ourse-service
PS C:\Users\Sean\Desktop\SAR2\Lab2\services\course-service> node src/server.
js
Course Service running on http://localhost:4002
```

Terminal 3

```
PS C:\Users\Sean\Desktop\SAR2> cd C:\Users\Sean\Desktop\SAR2\Lab2\services\e
nrollment-service
PS C:\Users\Sean\Desktop\SAR2\Lab2\services\enrollment-service> node src/ser
ver.js
Enrollment Service running on http://localhost:4003
Student service URL: http://localhost:4001
Course service URL: http://localhost:4002
```

C. Sequential Test Flow

Use Terminal 4 for all curl commands. Execute steps in order — do not skip steps.

Step 1 - GET all students (200)

```
PS C:\Users\Sean\Desktop\SAR2\Lab2> curl.exe -i http://localhost:4001/students | Tee-Object .\docs\evidence\01-get-students-200.txt
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
d
  0     0    0     0    0     0      0      0  --:--:-- --:--:-- --:--:--
100  490  100  490    0     0 18505      0  --:--:-- --:--:-- --:--:-- 18846
HTTP/1.1 200 OK
X-Powered-By: Express
Access-Control-Allow-Origin: *
Content-Type: application/json; charset=utf-8
Content-Length: 490
ETag: W/"1ea-eC3GIyeoNpS3YI+NsPdxtB/67E8"
Date: Fri, 20 Mar 2026 14:20:21 GMT
Connection: keep-alive
Keep-Alive: timeout=5

[{"id":4,"fullName":"Carla Andura","email":"carla.andura.bsit3b@example.com","createdAt":"2026-03-20T14:19:07.961Z"}, {"id":3,"fullName":"Kate Nicole Bermejo","email":"kate.bermejo.bsit3b@example.com","createdAt":"2026-03-20T14:19:07.961Z"}, {"id":2,"fullName":"Sophia Christi Garcia","email":"sophia.garcia.bsit3b@example.com","createdAt":"2026-03-20T14:19:07.961Z"}, {"id":1,"fullName":"Mariene Labrador","email":"mariene.labrador.bsit3b@example.com","createdAt":"2026-03-20T14:19:07.961Z"}]
PS C:\Users\Sean\Desktop\SAR2\Lab2>
```

Step 2 - GET all courses (200)

```
Connection: keep-alive ...
PS C:\Users\Sean\Desktop\SAR2\Lab2> curl.exe -i http://localhost:4002/courses | Tee-Object .\docs\evidence\02-get-course
s-200.txt
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload  Total   Spent    Left   Speed
100  613  100  613    0    0  23123      0 --:--:-- --:--:-- --:--:-- 23576
HTTP/1.1 200 OK
X-Powered-By: Express
Access-Control-Allow-Origin: *
Content-Type: application/json; charset=utf-8
Content-Length: 613
ETag: W/"265-0ruQc7cqr3lynFc5ZSNdMCnHdWM"
Date: Fri, 20 Mar 2026 14:25:50 GMT
Connection: keep-alive
Keep-Alive: timeout=5

[{"id":6,"code":"BSIT3B-SAR2-B","title":"SAR2 Microservices B","createdAt":"2026-03-20T14:25:17.174Z"}, {"id":5,"code":"BSIT3B-SAR2-A","title":"SAR2 Microservices A","createdAt":"2026-03-20T14:25:17.174Z"}, {"id":4,"code":"BSIT3B-SAR2-B","title":"SAR2 Microservices B","createdAt":"2026-03-20T14:24:45.198Z"}, {"id":3,"code":"BSIT3B-SAR2-A","title":"SAR2 Microservices A","createdAt":"2026-03-20T14:24:45.198Z"}, {"id":2,"code":"BSIT3B-SAR2-B","title":"SAR2 Microservices B","createdAt":"2026-03-20T14:23:12.200Z"}, {"id":1,"code":"BSIT3B-SAR2-A","title":"SAR2 Microservices A","createdAt":"2026-03-20T14:23:12.200Z"}]
```

Step 3 - POST student (201)

```
PS C:\Users\Sean\Desktop\SAR2\Lab2> '{"fullName":"Example Student","email":"example.student.seq@example.com"}' | curl.exe -i -X POST "http://localhost:4001/students" -H "Content-Type: application/json" --data-binary "@-" | Tee-Object .\docs\evidence\03-post-student-201.txt
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload  Total   Spent    Left   Speed
100  193  100  119  100   74   6074   3777 --:--:-- --:--:-- --:--:-- 10157
HTTP/1.1 201 Created
X-Powered-By: Express
Access-Control-Allow-Origin: *
Content-Type: application/json; charset=utf-8
Content-Length: 119
ETag: W/"77-WgFU5+1d1JyuV8IZUHlGPdP1jnE"
Date: Fri, 20 Mar 2026 14:33:59 GMT
Connection: keep-alive
Keep-Alive: timeout=5

{"id":11,"fullName":"Example Student","email":"example.student.seq@example.com","createdAt":"2026-03-20T14:33:59.901Z"}
```

Use the returned student id from Step 3 for Steps 4, 5, 25, and 27. If you get id 11 in Step 3, replace /students/5 with /students/11 in those steps.

Step 4 - GET created student by ID (200)

```
PS C:\Users\Sean\Desktop\SAR2\Lab2> $studentId = (Get-Content .\docs\evidence\03-post-student-201.txt -Raw | Select-String -Pattern '"id":(\d+)' | ForEach-Object { $_.Matches[0].Groups[1].Value })
PS C:\Users\Sean\Desktop\SAR2\Lab2> curl.exe -i "http://localhost:4001/students/$studentId" | Tee-Object .\docs\evidence\04-get-student-200.txt
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
100  118  100  118    0     0   5974      0 --:--:-- --:--:-- --:--:--  6210
HTTP/1.1 200 OK
X-Powered-By: Express
Access-Control-Allow-Origin: *
Content-Type: application/json; charset=utf-8
Content-Length: 118
ETag: W/"76-PfId0FwFFDkCNfWSX+1GvY+uHg4"
Date: Sun, 22 Mar 2026 12:09:37 GMT
Connection: keep-alive
Keep-Alive: timeout=5

{"id":5,"fullName":"Example Student","email":"example.student.seq@example.com","createdAt":"2026-03-22T12:09:20.370Z"}
```

Step 5 - PUT update student (200)

```
PS C:\Users\Sean\Desktop\SAR2\Lab2> '{"fullName":"Example Student Updated","email":"example.student.updated@example.com"}' | curl.exe -i -X PUT "http://localhost:4001/students/$studentId" -H "Content-Type: application/json" --data-binary "@-" | Tee-Object .\docs\evidence\05-put-student-200.txt
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
100  216  100  216    0     0   5312   3514 --:--:-- --:--:-- --:--:--  9000
HTTP/1.1 200 OK
X-Powered-By: Express
Access-Control-Allow-Origin: *
Content-Type: application/json; charset=utf-8
Content-Length: 130
ETag: W/"82-rStPIeSej3SHPKdjW5vUJz5V6PB8"
Date: Sun, 22 Mar 2026 12:10:01 GMT
Connection: keep-alive
Keep-Alive: timeout=5

{"id":5,"fullName":"Example Student Updated","email":"example.student.updated@example.com","createdAt":"2026-03-22T12:09:20.370Z"}
```

Step 6 - POST course (201)

```

PS C:\Users\Sean\Desktop\SAR2\Lab2> '{"code":"ITSAR2-SEQ","title":"Sequential Course"}' | cu
rl.exe -i -X POST "http://localhost:4002/courses" -H "Content-Type: application/json" --data
-binary "@-" | Tee-Object .\docs\evidence\06-post-course-201.txt
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload  Total   Spent    Left   Speed
100    146    100    95    100     51   1797    964  --:--:-- --:--:-- --:--:--   2754
HTTP/1.1 201 Created
X-Powered-By: Express
Access-Control-Allow-Origin: *
Content-Type: application/json; charset=utf-8
Content-Length: 95
ETag: W/"5f-fqvkr1X2lq5pjSIpd80YDU0UrLQ"
Date: Sun, 22 Mar 2026 12:11:10 GMT
Connection: keep-alive
Keep-Alive: timeout=5

{"id":3,"code":"ITSAR2-SEQ","title":"Sequential Course","createdAt":"2026-03-22T12:11:10.131Z"}

```

Step 7 - GET created course by ID (200)

```

PS C:\Users\Sean\Desktop\SAR2\Lab2> curl.exe -i http://localhost:4002/courses/3 | Tee-Object
.\docs\evidence\07-get-course-200.txt
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload  Total   Spent    Left   Speed
100     95    100     95     0     0   4898     0  --:--:-- --:--:-- --:--:--   5000
HTTP/1.1 200 OK
X-Powered-By: Express
Access-Control-Allow-Origin: *
Content-Type: application/json; charset=utf-8
Content-Length: 95
ETag: W/"5f-fqvkr1X2lq5pjSIpd80YDU0UrLQ"
Date: Sun, 22 Mar 2026 12:11:26 GMT
Connection: keep-alive
Keep-Alive: timeout=5

{"id":3,"code":"ITSAR2-SEQ","title":"Sequential Course","createdAt":"2026-03-22T12:11:10.131Z"}

```


Step 8 - PUT update course (200)

```
    '{"code":"ITSAR2-SEQ-UPD","title":"Sequential Course Upd
● ated"}' | curl.exe -i -X PUT "http://localhost:4002/courses/3" -H "Content-Type: application
/json" --data-binary "@-" | Tee-Object .\docs\evidence\08-put-course-200.txt
43e3-a500-7bdc25
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total     Spent    Left  Speed
100  170  100   107  100    63   4281   2520  --:--:-- --:--:-- --:--:--  7083
HTTP/1.1 200 OK
X-Powered-By: Express
Access-Control-Allow-Origin: *
Content-Type: application/json; charset=utf-8
Content-Length: 107
ETag: W/"6b-ije76PY0gmpn4Ru02/axo8inK0k"
Date: Sun, 22 Mar 2026 12:11:52 GMT
Connection: keep-alive
Keep-Alive: timeout=5

{"id":3,"code":"ITSAR2-SEQ-UPD","title":"Sequential Course Updated","createdAt":"2026-03-22T
12:11:10.131Z"}
○ PS C:\Users\Sean\Desktop\SAR2\Lab2>
```

Step 9 - POST enrollment (201)

```
PS C:\Users\Sean\Desktop\SAR2\Lab2> '{"studentId":1,"courseId":1}' | curl.exe -i -X POST "ht
● tp://localhost:4003/enrollments" -H "Content-Type: application/json" --data-binary "@-" | Te
e-Object .\docs\evidence\09-post-enrollment-201.txt
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total     Spent    Left  Speed
100  175  100   145  100    30   1585   328  --:--:-- --:--:-- --:--:--  1923
HTTP/1.1 201 Created
X-Powered-By: Express
Access-Control-Allow-Origin: *
Content-Type: application/json; charset=utf-8
Content-Length: 145
ETag: W/"91-iPM3vrtudU+BD8mUSzk8Fy6vXxE"
Date: Sun, 22 Mar 2026 12:12:09 GMT
Connection: keep-alive
Keep-Alive: timeout=5

{"id":3,"studentId":1,"courseId":1,"enrolledAt":"2026-03-22T12:12:09.507Z","studentName":"Ma
riene Labrador","courseTitle":"SAR2 Microservices A"}
```

Step 10 - GET enrollment by ID (200)

```
PS C:\Users\Sean\Desktop\SAR2\Lab2> curl.exe -i http://localhost:4003/enrollments/3 | Tee-Object .\docs\evidence\10-get-enrollment-200.txt
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total     Spent    Left     Speed
100  145  100  145    0     0  4817      0 --:--:-- --:--:-- --:--:-- 4833
HTTP/1.1 200 OK
X-Powered-By: Express
Access-Control-Allow-Origin: *
Content-Type: application/json; charset=utf-8
Content-Length: 145
ETag: W/"91-iPM3vrtudU+BD8mUSzk8Fy6vXxE"
Date: Sun, 22 Mar 2026 12:12:23 GMT
Connection: keep-alive
Keep-Alive: timeout=5

{"id":3,"studentId":1,"courseId":1,"enrolledAt":"2026-03-22T12:12:09.507Z","studentName":"M
ariene Labrador","courseTitle":"SAR2 Microservices A"}
```

Step 11 - GET enrollments list (200)

```
PS C:\Users\Sean\Desktop\SAR2\Lab2> curl.exe -i http://localhost:4003/enrollments | Tee-Object .\docs\evidence\11-get-enrollments-200.txt
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total     Spent    Left     Speed
100  447  100  447    0     0 12339      0 --:--:-- --:--:-- --:--:-- 12416
HTTP/1.1 200 OK
X-Powered-By: Express
Access-Control-Allow-Origin: *
Content-Type: application/json; charset=utf-8
Content-Length: 447
ETag: W/"1bf-1A7wS+LFerdu/STCZEYR6T4FbM0"
Date: Sun, 22 Mar 2026 12:12:42 GMT
Connection: keep-alive
Keep-Alive: timeout=5

[{"id":3,"studentId":1,"courseId":1,"enrolledAt":"2026-03-22T12:12:09.507Z","studentName":"M
ariene Labrador","courseTitle":"SAR2 Microservices A"}, {"id":2,"studentId":3,"courseId":2,"e
nrolledAt":"2026-03-22T12:08:23.010Z","studentName":"Kate Nicole Bermejo","courseTitle":"SAR
2 Microservices B"}, {"id":1,"studentId":2,"courseId":1,"enrolledAt":"2026-03-22T12:08:23.010
Z","studentName":"Sophia Christi Garcia","courseTitle":"SAR2 Microservices A"}]
```

Step 12 - Validation: POST student missing fullName (400)

```
PS C:\Users\Sean\Desktop\SAR2\Lab2> '{"email":"missing-name@example.com"}' | curl.exe -i -X POST "http://localhost:4001/students" -H "Content-Type: application/json" --data-binary "@-" | Tee-Object .\docs\evidence\12-student-missing-name-400.txt
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total     Spent    Left     Speed
100    81  100    43  100    38    2189    1935  --:--:-- --:--:-- --:--:--   4263
HTTP/1.1 400 Bad Request
X-Powered-By: Express
Access-Control-Allow-Origin: *
Content-Type: application/json; charset=utf-8
Content-Length: 43
ETag: W/"2b-nbhBQ9aPvpwCM8evMmtzRtNQ86g"
Date: Sun, 22 Mar 2026 12:13:00 GMT
Connection: keep-alive
Keep-Alive: timeout=5

{"error":"fullName and email are required"}
```

Step 13 - Validation: POST student missing email (400)

```
PS C:\Users\Sean\Desktop\SAR2\Lab2> '{"fullName":"No Email"}' | curl.exe -i -X POST "http://localhost:4001/students" -H "Content-Type: application/json" --data-binary "@-" | Tee-Object .\docs\evidence\13-student-missing-email-400.txt
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total     Spent    Left     Speed
100    68  100    43  100    25    2232    1298  --:--:-- --:--:-- --:--:--   3578
HTTP/1.1 400 Bad Request
X-Powered-By: Express
Access-Control-Allow-Origin: *
Content-Type: application/json; charset=utf-8
Content-Length: 43
ETag: W/"2b-nbhBQ9aPvpwCM8evMmtzRtNQ86g"
Date: Sun, 22 Mar 2026 12:13:16 GMT
Connection: keep-alive
Keep-Alive: timeout=5

{"error":"fullName and email are required"}
```


Step 14 - Validation: POST enrollment missing courseId (400)

```
PS C:\Users\Sean\Desktop\SAR2\Lab2> '{"studentId":1}' | curl.exe -i -X POST "http://localhost:4003/enrollments" -H "Content-Type: application/json" --data-binary "@-" | Tee-Object .\docs\evidence\14-enrollment-missing-course-400.txt
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           %    100    47   100      17    2400     868   --:--:--  --:--:--  --:--:--  3368
HTTP/1.1 400 Bad Request
X-Powered-By: Express
Access-Control-Allow-Origin: *
Content-Type: application/json; charset=utf-8
Content-Length: 47
ETag: W/"2f-CHQm9NCB/mTp20qGqbHLvZ6oYi4"
Date: Sun, 22 Mar 2026 12:13:26 GMT
Connection: keep-alive
Keep-Alive: timeout=5

{"error":"studentId and courseId are required"}
```

Step 15 - Not Found: GET nonexistent student (404)

```
PS C:\Users\Sean\Desktop\SAR2\Lab2> curl.exe -i http://localhost:4001/students/9999 | Tee-Ob
ject .\docs\evidence\15-student-notfound-404.txt
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
100    29  100    29    0    0   1421      0  --:--:-- --:--:-- --:--:--   1450
HTTP/1.1 404 Not Found
X-Powered-By: Express
Access-Control-Allow-Origin: *
Content-Type: application/json; charset=utf-8
Content-Length: 29
ETag: W/"1d-/w9leX+CSVwScG3mclc8pAqpxGg"
Date: Sun, 22 Mar 2026 12:14:03 GMT
Connection: keep-alive
Keep-Alive: timeout=5

{"error":"Student not found"}
```

Step 16 - Not Found: GET nonexistent course (404)

```
PS C:\Users\Sean\Desktop\SAR2\Lab2> curl.exe -i http://localhost:4002/courses/9999 | Tee-Object .\docs\evidence\16-course-notfound-404.txt
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           % Dload  % Upload   Total   Spent    Left     Speed
100    28    100    28    0    0   1460      0  --:--:-- --:--:-- --:--:--  1473
HTTP/1.1 404 Not Found
X-Powered-By: Express
Access-Control-Allow-Origin: *
Content-Type: application/json; charset=utf-8
Content-Length: 28
ETag: W/"1c-12dfk81SPpEzTq0cIIa3iq/Kl1Q"
Date: Sun, 22 Mar 2026 12:14:36 GMT
Connection: keep-alive
Keep-Alive: timeout=5

{"error":"Course not found"}
```

Step 17 - Not Found: GET nonexistent enrollment (404)

```
curl.exe -i http://localhost:4003/enrollments/9999 | Tee-Object .\docs\evidence\17-enrollment-notfound-404.txt
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           % Dload  % Upload   Total   Spent    Left     Speed
100    32    100    32    0    0   1757      0  --:--:-- --:--:-- --:--:--  1777
HTTP/1.1 404 Not Found
X-Powered-By: Express
Access-Control-Allow-Origin: *
Content-Type: application/json; charset=utf-8
Content-Length: 32
ETag: W/"20-K1Lj0fCQ90HKLExvRggmXZT9Np4"
Date: Sun, 22 Mar 2026 12:15:58 GMT
Connection: keep-alive
Keep-Alive: timeout=5

{"error":"Enrollment not found"}
```


Step 18 - Duplicate: POST student with existing email (409)

```
PS C:\Users\Sean\Desktop\SAR2\Lab2> '{"fullName":"Duplicate Student","email":"example.student.updated@example.com"}' | curl.exe -i -X POST "http://localhost:4001/students" -H "Content-Type: application/json" --data-binary "@-" | Tee-Object .\docs\evidence\18-student-duplicate-409.txt
```

% Total	% Received	% Xferd	Average Speed		Time	Time	Time	Current
			Dload	Upload	Total	Spent	Left	Speed
100	120	100	40	100	80	2026	4053	6315

```
HTTP/1.1 409 Conflict
X-Powered-By: Express
Access-Control-Allow-Origin: *
Content-Type: application/json; charset=utf-8
Content-Length: 40
ETag: W/"28-GPow/wop01VlG3Z59Zabhw+4lWU"
Date: Sun, 22 Mar 2026 12:16:13 GMT
Connection: keep-alive
Keep-Alive: timeout=5

{"error":"Student email already exists"}
```

Step 19 - Duplicate: POST enrollment already enrolled (409)

```
PS C:\Users\Sean\Desktop\SAR2\Lab2> '{"studentId":1,"courseId":1}' | curl.exe -i -X POST "http://localhost:4003/enrollments" -H "Content-Type: application/json" --data-binary "@-" | Tee-Object .\docs\evidence\19-enrollment-duplicate-409.txt
```

% Total	% Received	% Xferd	Average Speed		Time	Time	Time	Current
			Dload	Upload	Total	Spent	Left	Speed
100	84	100	54	100	30	2042	1134	3230

```
HTTP/1.1 409 Conflict
X-Powered-By: Express
Access-Control-Allow-Origin: *
Content-Type: application/json; charset=utf-8
Content-Length: 54
ETag: W/"36-mRsRene6V37I/5odgUY4K5nG8lM"
Date: Sun, 22 Mar 2026 12:16:29 GMT
Connection: keep-alive
Keep-Alive: timeout=5

{"error":"Student is already enrolled in this course"}
```

Step 20 - Cross-service Not Found: nonexistent studentId (404)

```
    '{"studentId":9999,"courseId":1}' | curl.exe -i -X POST
• "http://localhost:4003/enrollments" -H "Content-Type: application/json" --data-binary "@-" |
  Tee-Object .\docs\evidence\20-cross-student-404.txt
    % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                   Dload  Upload   Total   Spent    Left   Speed
 100    67  100    34  100    33   1492   1448  --:--:-- --:--:-- --:--:--  3045
HTTP/1.1 404 Not Found
X-Powered-By: Express
Access-Control-Allow-Origin: *
Content-Type: application/json; charset=utf-8
Content-Length: 34
ETag: W/"22-D3vra6urJG+TrASPrk+YgRzlvOw"
Date: Sun, 22 Mar 2026 12:16:56 GMT
Connection: keep-alive
Keep-Alive: timeout=5

{"error":"Student does not exist"}
```

Step 21 - Cross-service Not Found: nonexistent courseId (404)

```
PS C:\Users\Sean\Desktop\SAR2\Lab2> '{"studentId":1,"courseId":9999}' | curl.exe -i -X POST
• "http://localhost:4003/enrollments" -H "Content-Type: application/json" --data-binary "@-" |
  Tee-Object .\docs\evidence\21-cross-course-404.txt
    % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                   Dload  Upload   Total   Spent    Left   Speed
 100    66  100    33  100    33   1255   1255  --:--:-- --:--:-- --:--:--  2538
HTTP/1.1 404 Not Found
X-Powered-By: Express
Access-Control-Allow-Origin: *
Content-Type: application/json; charset=utf-8
Content-Length: 33
ETag: W/"21-XjsnrivIm8axEbPg9WLL7LQCZaA"
Date: Sun, 22 Mar 2026 12:17:12 GMT
Connection: keep-alive
Keep-Alive: timeout=5

{"error":"Course does not exist"}
```

Step 22 - Dependency Down simulation

Terminal 5:

```
PS C:\Users\Sean\Desktop\SAR2\Lab2> cd C:\Users\Sean\Desktop\SAR2\Lab2\services\enrollment-service
PS C:\Users\Sean\Desktop\SAR2\Lab2\services\enrollment-service> $env:PORT="4113"
PS C:\Users\Sean\Desktop\SAR2\Lab2\services\enrollment-service> $env:STUDENT_SERVICE_URL="http://localhost:4199"
PS C:\Users\Sean\Desktop\SAR2\Lab2\services\enrollment-service> $env:COURSE_SERVICE_URL="http://localhost:4002"
PS C:\Users\Sean\Desktop\SAR2\Lab2\services\enrollment-service> node src/server.js
Enrollment Service running on http://localhost:4113
Student service URL: http://localhost:4199
Course service URL: http://localhost:4002
Dependency timeout (ms): 2000
```

Terminal 4:

```
PS C:\Users\Sean\Desktop\SAR2\Lab2> '{"studentId":1,"courseId":1}' | curl.exe -i -X POST "http://localhost:4113/enrollments" -H "Content-Type: application/json" --data-binary "@-" | Tee-Object .\docs\evidence\22-dependency-down.txt
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total     Spent    Left     Speed
100    69    100    39    100    30     455    350  --:--:-- --:--:-- --:--:--    802
HTTP/1.1 503 Service Unavailable
X-Powered-By: Express
Access-Control-Allow-Origin: *
Content-Type: application/json; charset=utf-8
Content-Length: 39
ETag: W/"27-HMdoZLJ4MGHIdpmnA1sBAkhZqQI"
Date: Sun, 22 Mar 2026 12:17:53 GMT
Connection: keep-alive
Keep-Alive: timeout=5

{"error":"Student service unavailable"}
```

Expected after update: 503 Service Unavailable with error about student service unavailable.

Step 23 - Timeout simulation

Terminal 6:

No command needed. Keep the main student-service running on port 4001 in Terminal 1.

Terminal 7:

```
PS C:\Users\Sean\Desktop\SAR2> cd C:\Users\Sean\Desktop\SAR2\Lab2\services\enrollment-service
>> $env:PORT="4123"
>> $env:STUDENT_SERVICE_URL="http://localhost:4001"
>> $env:COURSE_SERVICE_URL="http://localhost:4002"
>> $env:DEPENDENCY_TIMEOUT_MS="1"
>> node src/server.js
src/server.js;ac99f3a9-5cae-4ca5-8484-67a8c91e9843Enrollment Service running on http://localhost:4123
Student service URL: http://localhost:4001
Course service URL: http://localhost:4002
Dependency timeout (ms): 1
```

Terminal 4:

```
PS C:\Users\Sean\Desktop\SAR2\Lab2> '{"studentId":1,"courseId":1}' | curl.exe -i -X POST "http://localhost:4123/enrollments" -H "Content-Type: application/json" --data-binary "@-" | Tee-Object .\docs\evidence\23-timeout.txt
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left   Speed
100    79  100    49  100    30     691    423  --:--:-- --:--:-- --:--:--  1128
HTTP/1.1 504 Gateway Timeout
X-Powered-By: Express
Access-Control-Allow-Origin: *
Content-Type: application/json; charset=utf-8
Content-Length: 49
ETag: W/"31-cFjpUNoIK7SsqgniFX0YgExt5Wg"
Date: Sun, 22 Mar 2026 12:18:57 GMT
Connection: keep-alive
Keep-Alive: timeout=5

{"error":"Student service unavailable (timeout)"}

```

Expected after update: 504 Gateway Timeout with error about student service unavailable (timeout).

Step 24 - DELETE enrollment by student (200)

```
PS C:\Users\Sean\Desktop\SAR2\Lab2> curl.exe -i -X DELETE http://localhost:4003/enrollments/by-student/1 | Tee-Object .\docs\evidence\24-delete-enrollment-200.txt
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload  Total  Spent    Left  Speed
100   13    100   13    0    0    514      0  --:--:-- --:--:-- --:--:--   520
HTTP/1.1 200 OK
X-Powered-By: Express
Access-Control-Allow-Origin: *
Content-Type: application/json; charset=utf-8
Content-Length: 13
ETag: W/"d-sxBY/LX4pXuaC7e8WH2aXAaM3wg"
Date: Sun, 22 Mar 2026 12:19:18 GMT
Connection: keep-alive
Keep-Alive: timeout=5

{"removed":1}
```

Step 25 - DELETE student (200)

```
PS C:\Users\Sean\Desktop\SAR2\Lab2> curl.exe -i -X DELETE http://localhost:4002/courses/3 | Tee-Object .\docs\evidence\26-delete-course-200.txt
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload  Total  Spent    Left  Speed
100   129    100   129    0    0   2516      0  --:--:-- --:--:-- --:--:--  2529
HTTP/1.1 200 OK
X-Powered-By: Express
Access-Control-Allow-Origin: *
Content-Type: application/json; charset=utf-8
Content-Length: 129
ETag: W/"81-EhgMD0MuWFI5ZEKTbqQRtjh8ECA"
Date: Sun, 22 Mar 2026 12:19:45 GMT
Connection: keep-alive
Keep-Alive: timeout=5

{"message":"Course deleted","course":{"id":3,"code":"ITSAR2-SEQ-UPD","title":"Sequential Course Updated"},"enrollmentsRemoved":0}
```

Step 26 - DELETE course (200)

```
PS C:\Users\Sean\Desktop\SAR2\Lab2> curl.exe -i -X DELETE http://localhost:4002/courses/3 | Tee-Object .\docs\evidence\26-delete-course-200.txt
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           %         0         0      Dload  Upload   Total   Spent    Left   Speed
100  129  100  129    0     0   2516      0 --:--:-- --:--:-- --:--:--  2529
HTTP/1.1 200 OK
X-Powered-By: Express
Access-Control-Allow-Origin: *
Content-Type: application/json; charset=utf-8
Content-Length: 129
ETag: W/"81-EhgMD0MuWFI5ZEKTbqQRtih8ECA"
Date: Sun, 22 Mar 2026 12:19:45 GMT
Connection: keep-alive
Keep-Alive: timeout=5

{"message":"Course deleted","course":{"id":3,"code":"ITSAR2-SEQ-UPD","title":"Sequential Course Updated"},"enrollmentsRemoved":0}
```

Step 27 - Verify cleanup returns 404

```
PS C:\Users\Sean\Desktop\SAR2\Lab2> curl.exe -i "http://localhost:4001/students/$studentId"
| Tee-Object .\docs\evidence\27-verify-student-404.txt
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
100    29    100    29    0     0   1524      0  --:--:-- --:--:-- --:--:--  1611
HTTP/1.1 404 Not Found
X-Powered-By: Express
Access-Control-Allow-Origin: *
Content-Type: application/json; charset=utf-8
Content-Length: 29
ETag: W/"1d-/w9leX+CSWwScG3mc1c8pAqpxGg"
Date: Sun, 22 Mar 2026 12:20:37 GMT
Connection: keep-alive
Keep-Alive: timeout=5

{"error":"Student not found"}
```

```

PS C:\Users\Sean\Desktop\SAR2\Lab2> curl.exe -i http://localhost:4003/enrollments/3 | Tee-Object .\docs\evidence\27-verify-enrollment-404.txt
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total     Spent    Left     Speed
100    32  100    32    0     0   1521      0  --:--:-- --:--:-- --:--:--  1600
HTTP/1.1 404 Not Found
X-Powered-By: Express
Access-Control-Allow-Origin: *
Content-Type: application/json; charset=utf-8
Content-Length: 32
ETag: W/"20-K1Lj0fCQ90HKLExvRggmXZT9Np4"
Date: Sun, 22 Mar 2026 12:20:58 GMT
Connection: keep-alive
Keep-Alive: timeout=5

{"error":"Enrollment not found"}

```

Member Dummy Data (Our Team)

```

PS C:\Users\Sean\Desktop\SAR2\Lab2> curl.exe -i http://localhost:4002/courses/3 | Tee-Object .\docs\evidence\27-verify-course-404.txt
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total     Spent    Left     Speed
100    28  100    28    0     0   1061      0  --:--:-- --:--:-- --:--:--  1076
HTTP/1.1 404 Not Found
X-Powered-By: Express
Access-Control-Allow-Origin: *
Content-Type: application/json; charset=utf-8
Content-Length: 28
ETag: W/"1c-12dfk81SPpEzTq0cIIa3iq/K11Q"
Date: Sun, 22 Mar 2026 12:20:47 GMT
Connection: keep-alive
Keep-Alive: timeout=5

{"error":"Course not found"}

```

These records are now seeded automatically on service start. Running the commands below may return 409 duplicate responses.

```

'{"fullName":"Mariene Labrador","email":"mariene.labrador.bsit3b@example.com"}' | curl.exe -i -X POST "http://localhost:4001/students" -H "Content-Type: application/json" --data-binary "@"-
'{"fullName":"Sophia Christi Garcia","email":"sophia.garcia.bsit3b@example.com"}' | curl.exe -i -X POST "http://localhost:4001/students" -H "Content-Type: application/json" --data-binary "@"-
'{"fullName":"Kate Nicole Bermejo","email":"kate.bermejo.bsit3b@example.com"}' | curl.exe -i -X POST "http://localhost:4001/students" -H "Content-Type: application/json" --data-binary "@"-
'{"fullName":"Carla Andura","email":"carla.andura.bsit3b@example.com"}' | curl.exe -i -X POST "http://localhost:4001/students" -H "Content-Type: application/json" --data-binary "@"-

```


III. EXPERIMENTAL FINDINGS / OBSERVATIONS

All 27 test steps executed in sequential order returned the correct HTTP status codes in the Node.js/Express environment.

Steps 1 through 11 confirmed that all three services respond correctly to valid CRUD operations. The responses successfully returned JSON objects containing the `id`, `createdAt`, and relevant data fields.

Steps 12 through 14 confirmed that validation errors effectively catch missing parameters, returning HTTP 400 with specific messages such as `{"error": "fullName and email are required"}`. Steps 15 through 17 successfully demonstrated standard not-found handling, returning HTTP 404 with messages like `{"error": "Student not found"}`.

Steps 20 and 21 validated cross-service communication capabilities. The enrollment-service correctly queried the upstream student and course services, mapping their empty or failing responses to proper 404 errors (`{"error": "Student does not exist"}`).

Duplicate detection functioned flawlessly in Steps 18 and 19, halting duplicate data entry and throwing 409 Conflict errors.

The dependency failure tests in Steps 22 and 23 successfully demonstrated fault tolerance and proper error mapping. By using environment variables to point to a dead port, the service triggered a connection refusal and returned HTTP 503. By drastically lowering the timeout threshold (`DEPENDENCY_TIMEOUT_MS="1"`), the system triggered a timeout exception and correctly returned HTTP 504 `{"error": "Student service unavailable (timeout)"}`.

IV. DISCUSSIONS

Consistent error string contracts across all three Express services are essential for predictable frontend error handling. Because all three services reliably output the `{"error": "..."}` format when failing, client applications can easily parse and display these messages without requiring complex, service-specific logic.

Only the enrollment-service is capable of returning 503 and 504 errors because it is the only microservice in this architecture with external HTTP dependencies. When creating an enrollment, the service makes outbound network requests to verify both the student and the course. The student and course services only query their local SQLite databases and make no outbound calls; therefore, network timeouts and unavailable upstream dependencies are architecturally impossible for them.

The clever use of environment variables (`PORT`, `STUDENT_SERVICE_URL`, `DEPENDENCY_TIMEOUT_MS`) in PowerShell to simulate dependency failures (Steps 22 and 23) highlights the flexibility of Node.js microservices. Instead of modifying source code (like injecting `sleep()` functions), manipulating the environment allowed for rapid, configuration-driven edge case testing.

V. CONCLUSION

All 27 sequential test steps were executed via Windows PowerShell and confirmed to return the correct HTTP status codes and JSON error bodies. The test suite thoroughly covered every endpoint across the Node.js microservices: happy path CRUD, input validation errors, not-found scenarios, duplicate detection, cross-service communication, dependency failures, and resource cleanup.

Microservices introduce unique failure modes, specifically 503 Service Unavailable and 504 Gateway Timeout, that monolithic applications do not face. Testing confirmed that the enrollment-service—acting as the single point of cross-service communication—successfully implemented resilience handling to catch and format these network exceptions.

Future production improvements could include adding circuit breakers to prevent cascade failures when upstream services are down, implementing retry logic with exponential backoff for transient timeouts, and centralizing logs to trace requests as they jump between the isolated Express APIs.

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